

GLOBAL DEVELOPMENT & SOCIO-ECONOMIC RISK SEGMENTATION ANALYSIS

1. Problem Statement

The objective is to analyze country-level socio-economic data to identify risk levels, segment countries, and support aid allocation decisions.

2. Dataset Overview

Dataset includes indicators like income, GDP per capita, child mortality, life expectancy, inflation, exports, imports, and fertility rate.

3. Data Cleaning & Preprocessing

Handled missing values, removed duplicates, and treated outliers in income, GDP, and inflation.

4. Feature Engineering

Development Index, Trade Balance, and Health Impact Ratio were created to improve analysis quality.

5. Segmentation Logic

Countries are classified into segments like High Risk, Developed Nation, Emerging Economy, Health Critical, High Inflation Risk, and Low GDP Trap using rule-based conditions in SQL and Python.

6. EDA Insights

Income positively correlates with life expectancy. High child mortality is linked with low income and poor healthcare.

7. SQL Modeling

A structured country_stats table was created with CASE-based segmentation and analytical queries using GROUP BY.

8. Dashboard (Streamlit)

Three pages: Global Overview, Health & Economic Risk, and Segmentation Insights with SQL-based filtering.

9. Key Findings

Developed nations show high income and life expectancy. High-risk countries have low GDP and high child mortality.

10. Recommendations

Prioritize aid for high-risk countries, improve healthcare investment, and support economic stabilization programs.

11. Conclusion

This project provides a data-driven framework for global socio-economic analysis and aid distribution.